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1913-18

Logic Circuit (Old question solutions)

2020 Fall. Q.1.

(a) Define digital system. Why do you prefer digital systems against its analog counterpart? Explain.

=>

The system which works in digital signal is called digital system. It works for discrete values and have greater accuracy.

Digital system over analog system:

- Digital data can be transmitted more effiently and is more reliable than analog data.
- Digital data can be stored in memory.
- It is less affected from the noise.
- Digital systems are more versatile.
- More digital circuitry can be fabricated on single IC chips.
- The transmission is more secure with encryption.
- Discrete value is used
- Have greater accuracy
- Easy to read data.

(b) Perform the following substraction using (r-1)'s complement.

i) $(111001)_2 - (11011)_2$

ii) $(2321)_{10} - (8301)_{10}$

Sol (i) $(111001)_2 - (11011)_2$

(a) 1's or (r-1)'s complement of 011011 is 100100

(b) Addition,

$$\begin{array}{r} 111001 \\ +\ 100100 \\ \hline \underline{1}011101 \end{array}$$

(c) Carry is present so ans is +ve

Then true ans = 011101 + 1

= 11110

(ii) $(2321)_{10} - (8301)_{10}$

(a) (r-1)'s complement of 8301 = 1698

(b) Addition,

$$\begin{array}{r} \cancel{2321} \\ \cancel{+\ 8301} \\ \hline \cancel{10622} \end{array} \qquad \begin{array}{r} 2321 \\ +\ 1698 \\ \hline 4019 \end{array}$$

(c) Carry is not present so answer is negetive

True ans = (r-1)'s complement of sum

= $-(5980)_{10}$

A.

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(c) What is weighted and non-weighted code? Why gray code is called reflected code?

Sol - If the value of symbol depends on position where it lies then it is called weighted code. eg. Decimal, binary etc.

- If the value of symbol is independent of the position where it is lies then it is called non-weighted code. eg. gray code, excess 3 code etc.

Decimal	8421 code	Gray code
0	0000	0000
1	0001	0001
2	0010	0011
3	0011	0010
4	0100	0110
5	0101	0111
6	0110	0101
7	0111	0100
8	1000	1100
9	1001	1101
10	1010	1111
11	1011	1110
12	1100	1010
13	1101	1011
14	1110	1001
15	1111	1000

In gray code there is a difference of only one bit position on successive increment of each decimal number. So that it is called reflected code.

2019 spring Q.1.

(a) With characters differentiate between Digital and Analog system.

→)

Digital system	Analog system
1. Digital system uses discrete signal as on / off representing binary format. 0 and 1.	Analog system uses continuous signal with varying magnitude
2. Digital system uses square wave.	Analog system uses sine wave
3. First transform the analog waves to limited set of numbers and then record them as digital square waves.	Record the physical waveform as as they are orginally generated.
4. Digital transmission is easy and can be made noise proof with no loss at all.	Analog systems are affected badly by noise during transmission.
5. Digital system hardwere can be easily modulated as per the requirements.	Analog system's hardwere are not flexible.
6. Needs more bandwidth to carry same information.	Require less bandwidth
7. Data is stored in form of bits	Data is stored in form of waveform signal.
8. Consumes low power.	Consumes high power.
9. Digital system is costly	Analog systems are cheap.
10. eg. Computer, CD, DVD	eg. Analog electronics, voice radio using AM frequency.

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(b) Perform the conversion as indicated.

(i) $(543)_6 = (\quad)_{excess\text{-}3}$.

		Rem.
2	543	
2	271	1
2	135	1
2	67	1
2	33	1
2	18	1
2	8	0
2	4	0
2	2	0
2	1	0
	0	1

$\Rightarrow (543)_6 = (1000011111)_2$

$= (0010\ 0001\ 1111)_2$

8 4 2 1

$\Rightarrow$ 5 4 3

$6^2\ 6^1\ 6^0$

$\Rightarrow 180 + 24 + 3$

$\Rightarrow (207)_{10}$

$\Rightarrow (0010\ 0000\ 0111)_2$

$\Rightarrow (010100111010)_{excess\text{-}3}$.

(ii) $(708)_{10} = (\quad)_{2421}$

$\Rightarrow$ 708

$\Rightarrow (0111\ 0000\ 1110)_{2421}$.

(iii) $(BBA)_{16} = (\quad)_2$.

$(BBA)_{16} \Rightarrow (101110111010)_2$

11 11 10

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(C) Use 2's complement to subtract the following.

(i) $(1011)_2 - (10100)_2$

(a) 2's complement of 10100 = 01011 + 1 = 01100

(b)

$$\begin{array}{r} 01011 \\ +\ 01100 \\ \hline 10111 \end{array}$$

(c) carry is not present so ans is true.

~~Discarding carry and Ans => $(0111)_2$.~~

True answer = 2's complement of 10111

= 10111 + 1

= 11000

=> $-(11000)_2$.

(ii) $(952)_{10} - (873)_{10}$

→ 8's complement of 873 = 126 + 1 = 127

→ Addition,

$$\begin{array}{r} 952 \\ +\ 127 \\ \hline 1079 \end{array}$$

→ Carry is present so ans is true.

Discard carry and ans is => $(079)_{10}$

(iii) $(968)_{BCD} - (256)_{BCD}$

→ r's complement of 256 = 743 + 1 = 744

→ Addition,

$$\begin{array}{r} 968 \\ +\ 744 \\ \hline 1112 \end{array}$$

→ Carry is present so ans is true, Discard carry and

ans => $(112)_{BCD}$.

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2019 Fall Q.1.

(a) Explain the digital number system. List out the advantages of digital system over the analog system.
done!!

(b) "Excess-3 code is a self-complementing code". Justify your answer with appropriate illustration.

Ans:

Decimal	BCD code	Excess-3 code
0	0000	0011
1	0001	0100
2	0010	0101
3	0011	0110
4	0100	0111
5	0101	1000
6	0110	1001
7	0111	1010
8	1000	1011
9	1001	1100

From above table of Excess-3 code we can see that, 1's complement of excess-3 value of decimal zero is equal to excess-3 of decimal 9. Similarly excess-3 of decimal 1 is equal to excess-3 of 1's complement of decimal 8. and so one.
Hence Excess-3 code is also called self-complementing code.

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(c) Explain (r-1)'s complement with example.

(i) (r-1)'s complement of binary number.

- Also called 1's complement.
- 1's complement of binary number is obtained by inverting each symbol of binary number.
- It is used to subtract two binary number.

Eg.

If $(1010101)_2$ is binary number then its 1's complement is $(0101010)_2$.

(ii) (r-1)'s complement of decimal number.

- Also called 9's complement.
- It is obtained by differentiating each digit by 9.

Eg. $(4532)_{10}$

its (r-1)'s complement is $\Rightarrow (5467)_{10}$.

2018 spring. Q.1.

(b) perform the conversion as indicated.

(i) $(235)_6 = (\ \ )$ excess 3

$(235)_6 = 2\times6^2 + 3\times6^1 + 5\times6^0$

$= (95)_{10}$

$= (1001010 1)_2$

$= (11001000)$ excess-3.

(ii) $(369)_{10} = (\quad)_{2421}$ 8421

$(369)_{10} \Rightarrow (0011\ 0110\ 1111)_{2421}$.

(iii) $(BCA)_{16} = (\ )_2$

$(B\ C\ A)_{16} = (101111001010)_2$.

↑ ↑ ↑
11 12 10

(c) use 2's complement to substract the following.

(i) $(1010)_2 - (10100)_2$

2's complement of 10100
= 01011 + 1
= 01100

  01010
+ 01100
  10110

⇒ carry is not present so. ans is -ve.

Ans ⇒ 2's complement of 10110
= 01001 + 1
= 01010
⇒ $-(01010)_2$.

(ii) $(957)_{10} - (876)_{10}$

1's complement of 876 = 123 + 1
= 124

  957
+ 124
 1081

⇒ carry present ans is +ve and

Ans ⇒ $(081)_{10}$

(iii) $(378)_{BCD} - (256)_{BCD}$

1's complement of 256 = 743 + 1
= 744

  378
+ 744
 1122

⇒ carry is present so ans is +ve

Ans ⇒ $(122)_{BCD}$.

2018 Fall Q.1.

(b) Convert the following conversions:

(i) $(101001.101)_2 = (?)_{10}$

$\underline{0010}\,\underline{1001}.\underline{1010} \Rightarrow (29.10)_{10}$

2 9 10

(ii) $(ABD)_{16} = (?)_8$

$(ABD)_{16} = (101010111101)_2$

10 11 13 $= (5275)_8$.

(iii) $(10101101)_2 = (?)$ gray.

= 10101101

11111011 $\Rightarrow (11111011)$ gray.

(iv) $(175.351)_8 = (?)_{16}$

$(175.351)_8 = (001111101.011101001000)_2$

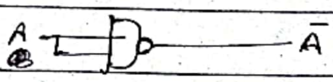
$\Rightarrow (7D.748)_{16}$.

(c) Define logic gates. Explain the universality of NAND and NOR gates.

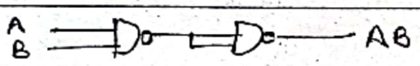
⇒ logic gates are the basic building block of any digital system which are used to perform any logic operation.

Universality of NAND gate:

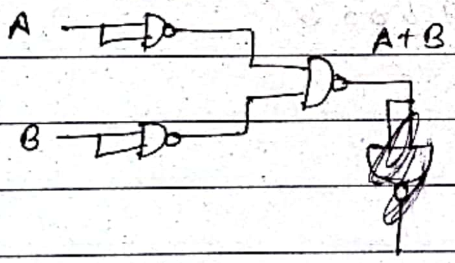
(a) As NOT gate



(b) As AND gate

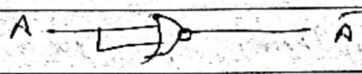


(c) As OR gate

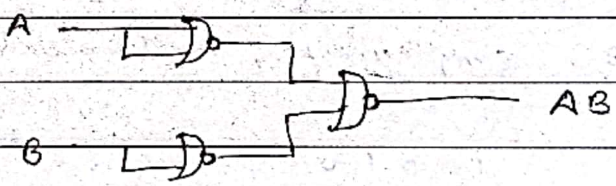


Universality of NOR gate:

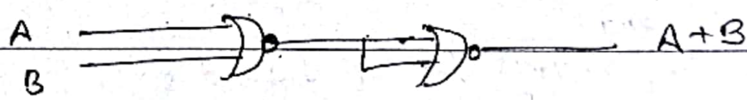
(a) As NOT gate



(b) As AND gate



(c) As OR gate



2017 spring Q.1.

(b) perform the following subtraction using 2's complement method.

(i) $(1000100)_2 - (1010100)_2$

2's complement of 1010100 ⇒ 0101011 +1

= 0101100

$$\begin{array}{r} 1000100 \\ +\ 0101100 \\ \hline 1110000 \end{array}$$

carry is not present so ans is -ve

2's complement of sum ⇒ 0001111 +1

⇒ 0010000

⇒ Ans = $-(0010000)_2$

(ii) $(11010)_2 - (10000)_2$

2's complement of 10000 = 01111 + 1

= 10000

$$\begin{array}{r} 11010 \\ +\ 10000 \\ \hline 101010 \end{array}$$

carry is present so ans is +ve.

and

Ans ⇒ $(01010)_2$.

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(c) What are the different types of Binary codes? Explain each in brief.

=) Different types of Binary codes are.

→ BCD codes
→ Excess-3 codes
→ 84-2-1 codes
→ 2421 codes

Decimal	BCD	Excess-3	84-2-1	2421
0	0000	0011	0000	0000
1	0001	0100	0111	0001
2	0010	0101	0110	0010
3	0011	0110	0101	0011
4	0100	0111	0100	0100
5	0101	1000	1011	0101
6	0110	1001	1010	0110
7	0111	1010	1001	0111
8	1000	1011	1000	1110
9	1001	1100	1111	1111

2017 Fall Q.1.

(b) Determine the value of base x if $(211)_x = (152)_8$

=) $2\times x^2 + 1\times x^1 + 1\times x^0 = 1\times 8^2 + 5\times 8^1 + 2\times 8^0$

=) $2x^2 + x + 1 = 106$

=) $2x^2 + x - 105 = 0$

solving,

$x = 7, -7.5$ ← not valid.

=) $x = 7$

2016 spring q.1.

(b) convert the following numbers from the given base to the other bases as indicated.

(i) $(11001.11)_2 = (\ )_8$

$(11001.11)_2 = (011001.110)_2$

$\Rightarrow (31.6)_8$.

(ii) $(5FE.DD)_{16} = (\ )_{10}$

$(5FE.DD)_{16} \Rightarrow 5\times16^2 + 15\times16^1 + 14\times16^0 + 13\times16^{-1} + 13\times16^{-2}$

$= (1534.863281)_{10}$.

(iii) $(777)_8 = (\ )_{16}$

$(777)_8 = (111\ 111\ 111)_2$

$= (0001\ 1111\ 1111)_2$

$= (1FF)_{16}$

(c) perform the subtraction with the following binary number using 1's complement

(i) 1010100 − 1000100

1's complement of 1000100 = 0111011

$$\begin{array}{r} 1010100 \\ +\ 0111011 \\ \hline 10001111 \end{array}$$

Carry is present so ans is +ve.

Add carry to LSP to get true ans $\Rightarrow (0010000)_2$

(ii) 1000100 − 1010100

1's complement of 1010100 = 0101011

```
  1000100
+ 0101011
---------
  1101111
```

no carry so ans is −ve.

1's complement of sum ⇒ 0010000

⇒ $-(10000)_2$ Ans.

2015 spring. Q.1.

(b) Perform the following conversion.

(i) $(573)_{10} = (\ )_{excess3}$.

⇒ $(573)_{10} = (0101\ 0111\ 0011)_2$

$= (1000\ 1010\ 0110)_{excess-3}$.

(ii) $(842)_{10} = (\ )_{2421}$

$(842)_{10} \Rightarrow (111001000010)_{2421}$

(iii) $(10101100)_{GRAY} = (\ )_2$.

```
10101100
1 + 0 0 0 1 0 0 0
```

⇒ $(11001000)_2$.

(iv) $(FAB)_{16} = (\ )_2$.

$(FAB)_{16} = (1111\ 1010\ 1011)_2$

15 10 11

2020 spring Q. 1.

simplify the given function F and don't care condition in SOP and POS form and draw the logic diagram with

1. NAND gate only
2. NOR gate only

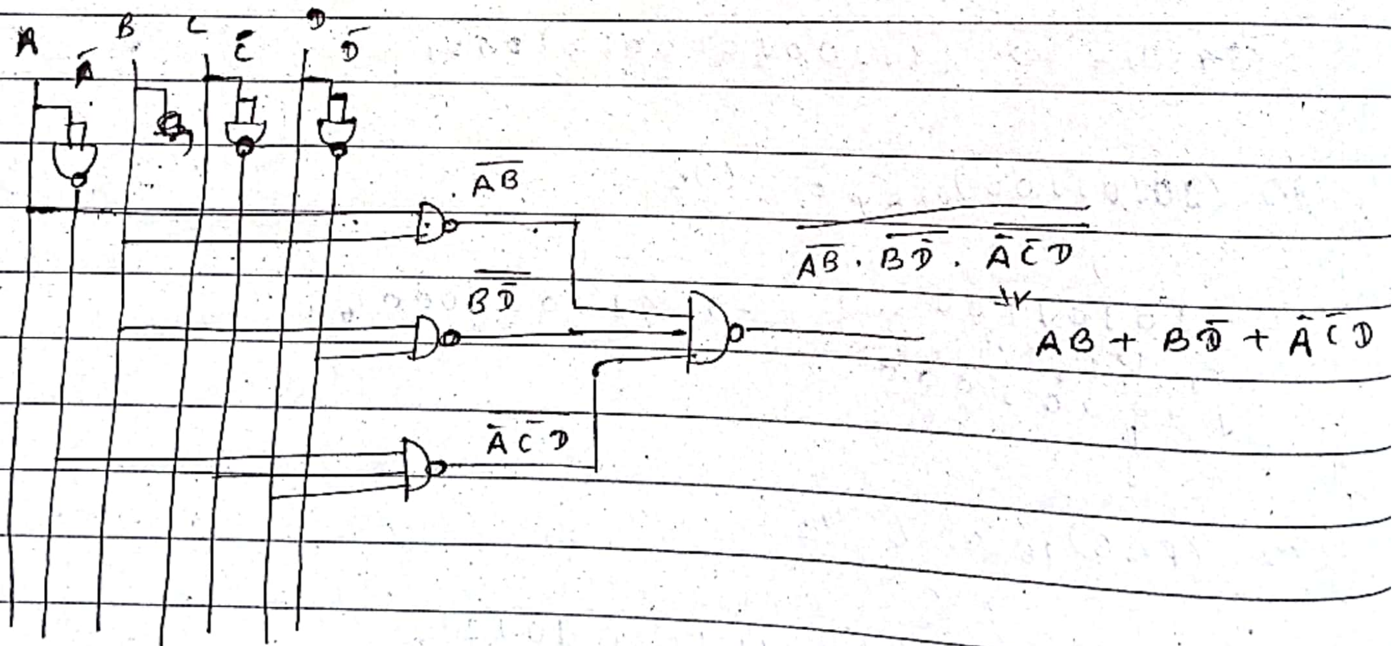
$F = \Sigma(1, 4, 5, 6, 12, 14, 15)$, $D = \Sigma(11, 13)$

(a) SOP form.

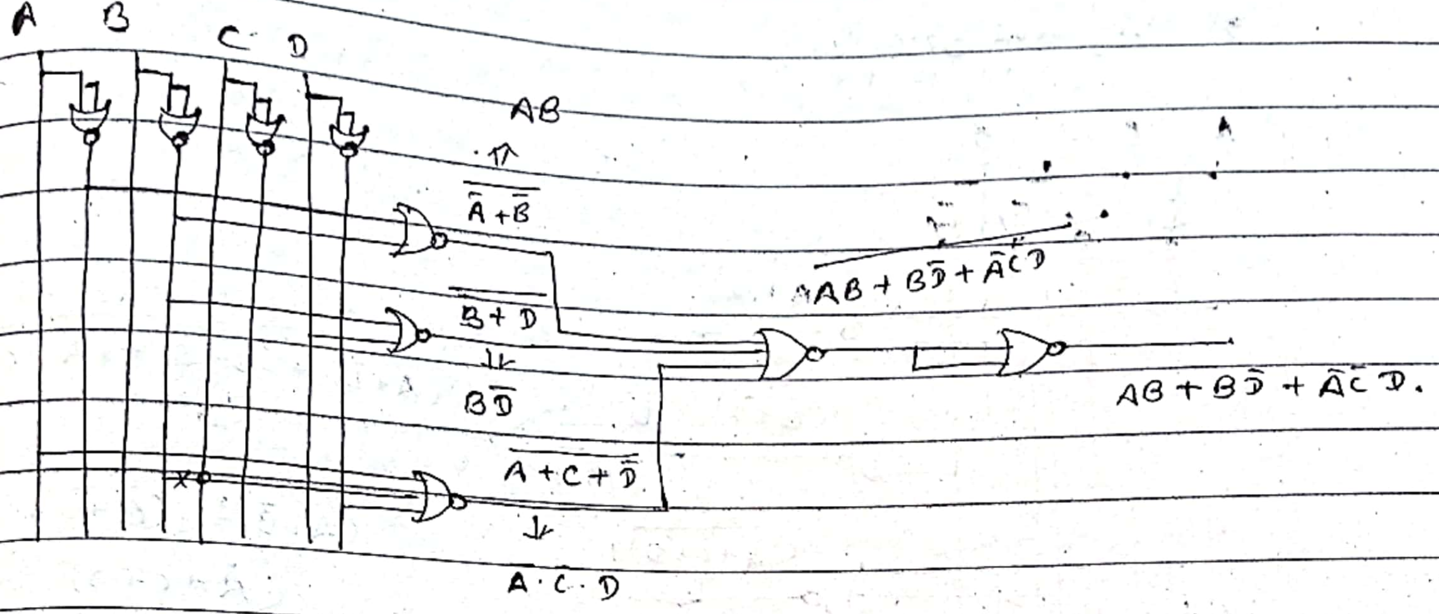
AB \ CD	00	01	11	10
00	0	1	0	0
01	1	1	0	1
11	1	x	1	1
10	0	0	x	0

$F = AB + B\bar{D} + \bar{A}\bar{C}D$

1. NAND gate only.



2. NOR gate only.

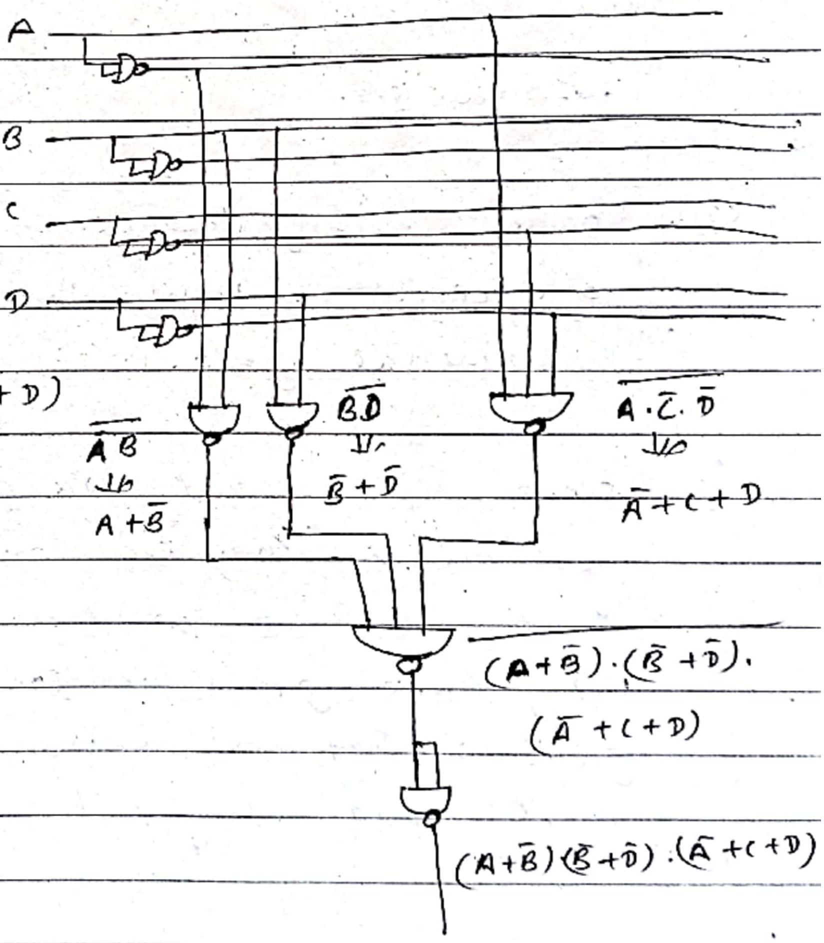


(b) POS form:

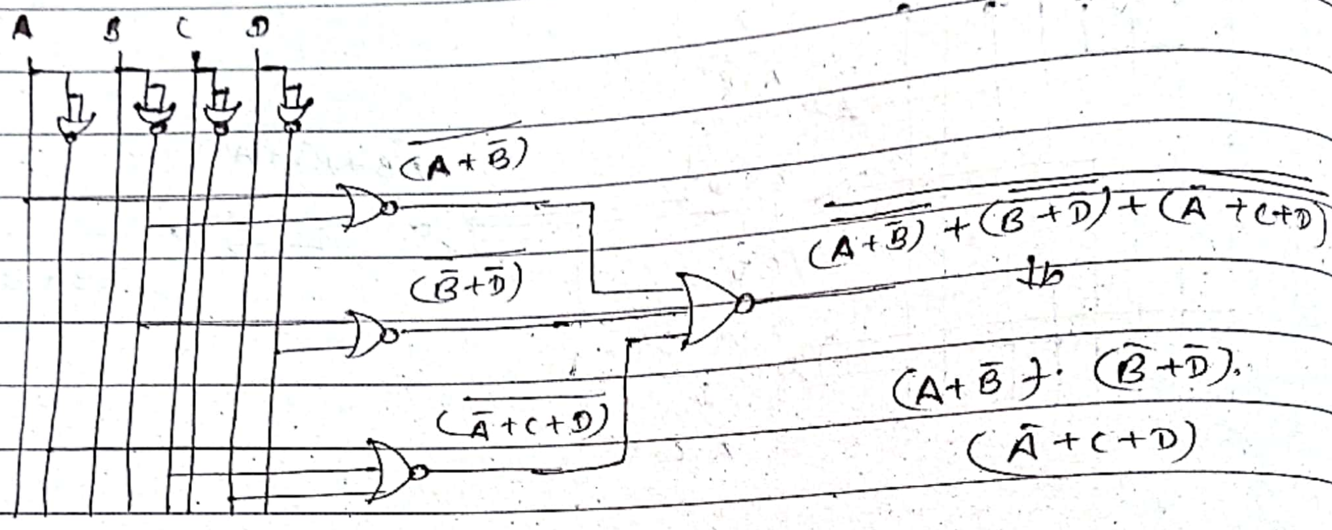
AB \ CD	00	01	11	10
00	0	1	0	0
01	1	1	0	1
11	1	x	1	1
10	0	0	x	0

$$F = (A+\bar{B})\cdot(\bar{B}+\bar{D})\cdot(\bar{A}+C+D)$$

1. NAND gate only.



9. Using NOR gate only.



$(A+\bar{B})\cdot(\bar{B}+\bar{D})\cdot(\bar{A}+C+D)$

2020 Fall Q-2.

(a) Discuss the significance of NAND and NOR gate in logic circuit. Prove that NAND and NOR gates are the universal gate.

⇒ Since NAND and NOR gates are the universal gates that means they can implement any boolean function without need to use any other gate type. Using these gates we can reduce the number of gates so they are economical and easier to fabricate in ICs and they used in all logic circuits.